

Credit Card Fraud Detection - Project Flow

This document describes the workflow followed to build the Credit Card Fraud Detection system using machine learning and Flask.

1. Environment Setup

Install Python, required libraries, and configure the development environment using Visual Studio Code.

2. Dataset Collection

Download the Credit Card Fraud Detection dataset and verify its structure.

3. Data Analysis

Import libraries, load the dataset, inspect features, and perform exploratory data analysis.

4. Data Preprocessing

Handle missing values (if any), remove duplicates, scale features, and prepare the data for model training.

5. Model Building

Train and evaluate machine learning models such as Logistic Regression, Decision Tree, and Random Forest. Select the best-performing model.

6. Model Saving

Save the trained machine learning model and scaler using Joblib.

7. Web Application Development

Develop a Flask application with an HTML interface to accept transaction details and predict whether a transaction is Legitimate or Fraudulent.

8. Testing and Validation

Test the application using sample legitimate and fraudulent transactions to verify correct predictions.

9. Deployment

Prepare the application for demonstration and GitHub submission.

Technologies Used

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Flask, HTML, CSS, Joblib, GitHub.